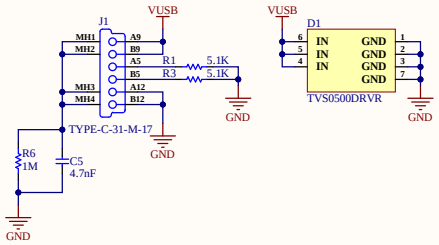
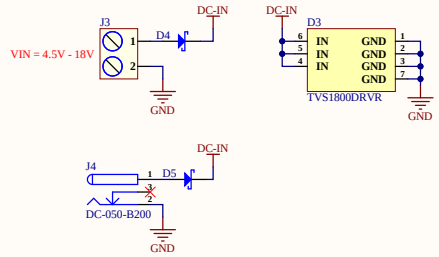
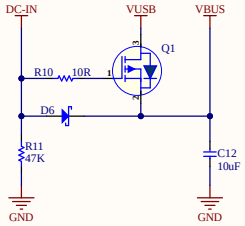
## USB INPUT



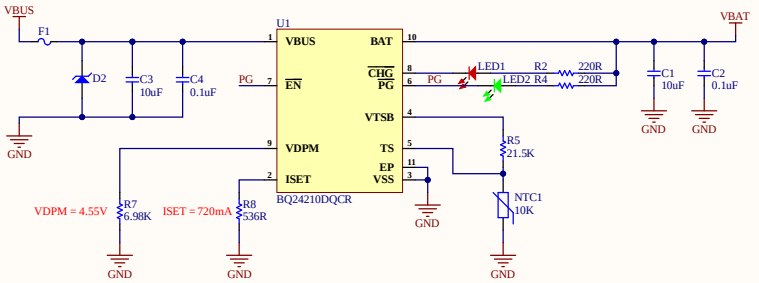

## DC OR SOLAR INPUT



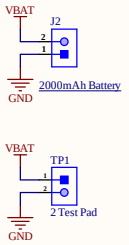

## POWER ORING



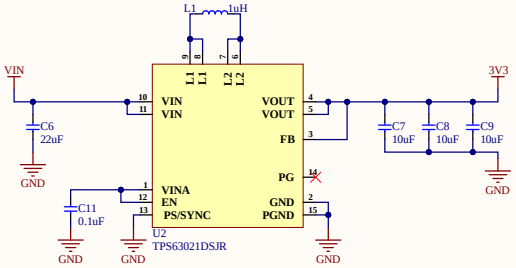

## Li-ION BATTERY CHARGER



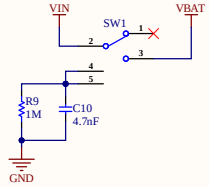

## BATTERY



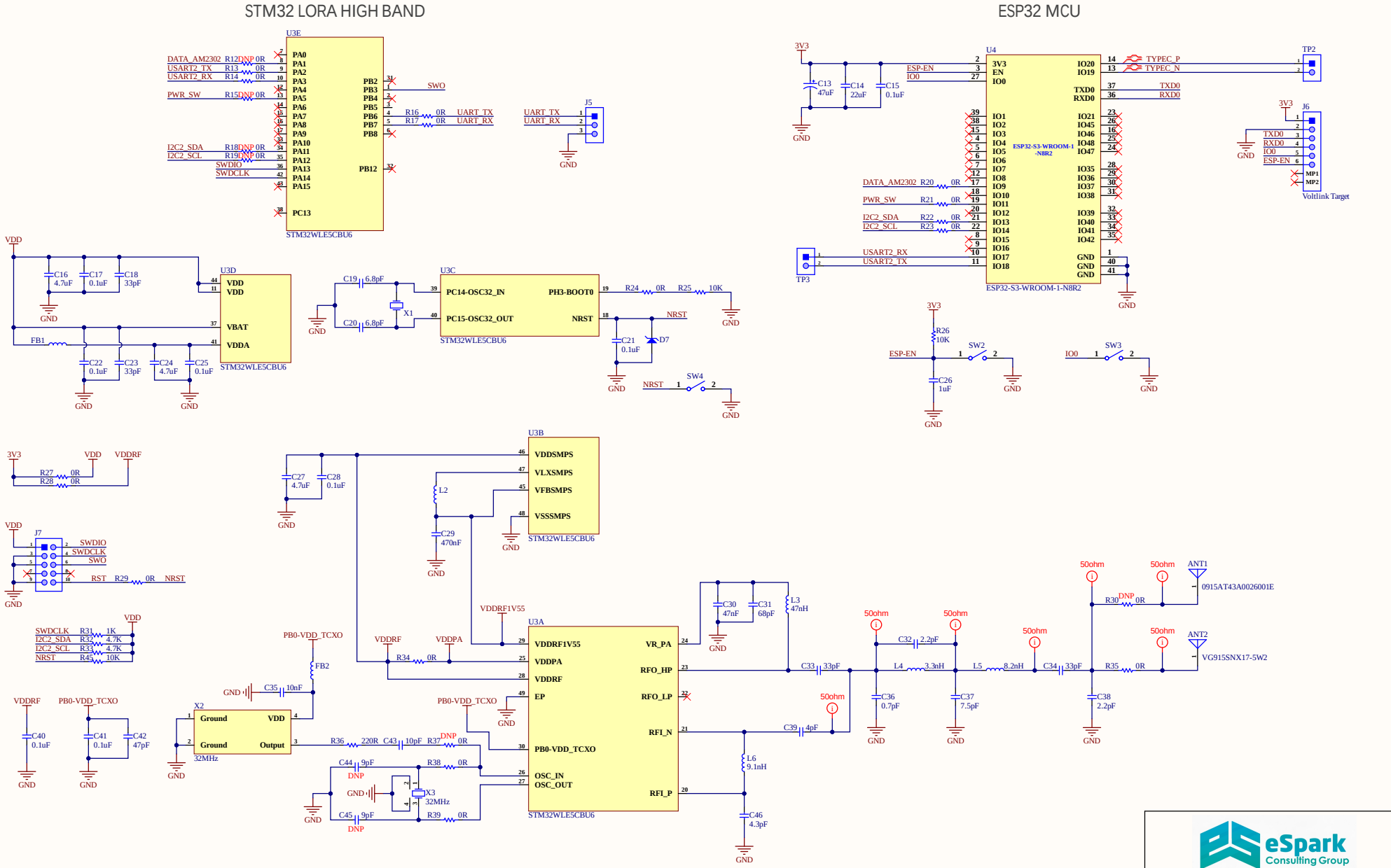

## BUCK - BOOST CONVERTER



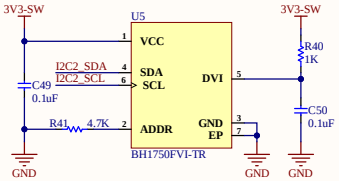

## SWITCH



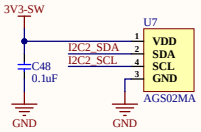

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| File: Gateway-Device-v1.0.PrjPcb | |
| Date: 7/8/2025 | Page Num: 1 |
| Drawn By: Huzaifa Ali | Page Total: 3 |


STM32 LORA HIGH BAND
ESP32 MCU
U3E
STM32WLE5CBU6
U4
ESP32-S3-WROOM-1-N8R2
U3D
U3C
U3B
U3A
J5
J6
J7
TP2
TP3
X1
X2
X3
32MHz
SW2
SW3
SW4
D7
FB1
FB2
3V3
VDD
VDDRF
GND
DATA_AM2302
USART2_TX
USART2_RX
PWR_SW
I2C2_SDA
I2C2_SCL
SWDIO
SWDCLK
SWO
NRST
RST
UART_TX
UART_RX
ESP-EN
IO0
TYPEC_P
TYPEC_N
TXD0
RXD0
VoltLink Target
ANT1
ANT2
0915AT43A0026001E
VG915SNX17-5W2
50ohm
PB0-VDD_TCXO
VDDRF1V55
VDDPA
DNP
eSpark Consulting Group
Title: MCU-Sheet.SchDoc
File: Gateway-Device-v1.0.PrjPcb
Date: 7/9/2025
Page Num: 2
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Page Total: 3

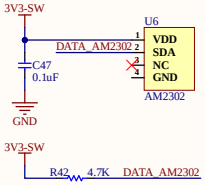
## LIGHT SENSOR



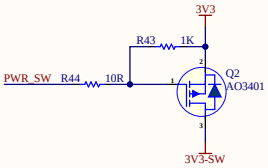

## AIR QUALITY SENSOR



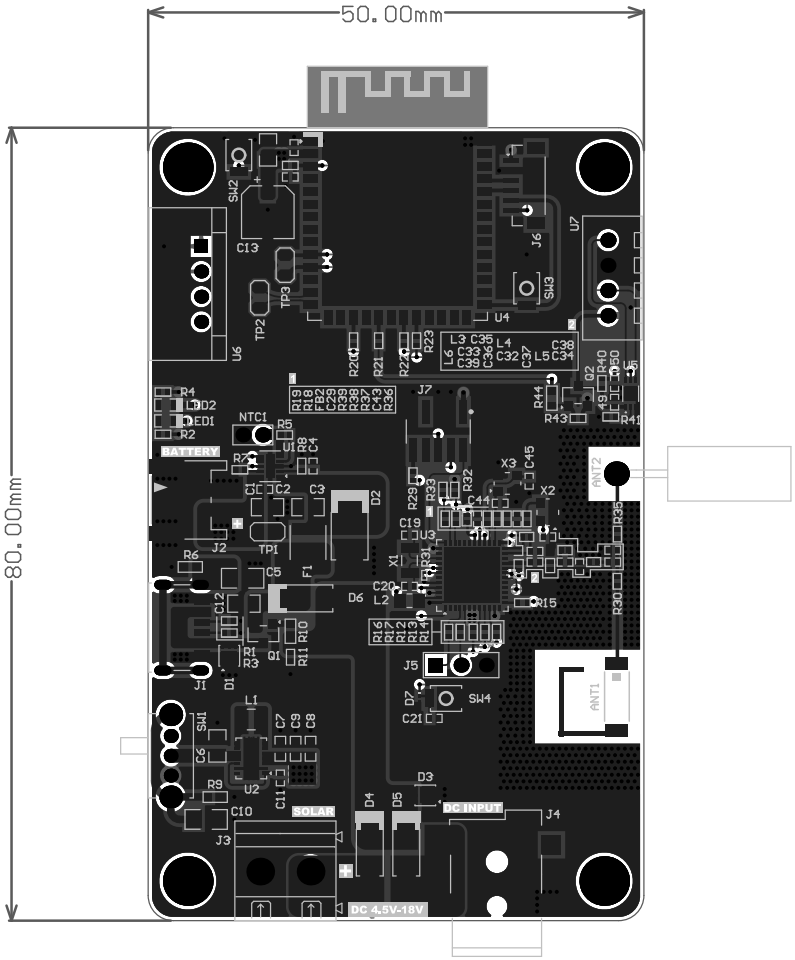

## TEMPERATURE AND HUMIDITY SENSOR




## HIGH SIDE SWITCH




| | |
|---|---|
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| File: Gateway-Device-v1.0.PrjPcb | |
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| Layer | Name | Material | Thickness | Constant | Board Layer Stack |
|---|---|---|---|---|---|
| | Top Overlay | | | | |
| | Top Solder | Solder Resist | 0.012mm | 3.5 | |
| 1 | Top Layer | | 0.035mm | | |
| | Dielectric 2 | PP-006 | 0.203mm | 4.4 | |
| 2 | Ground | CF-004 | 0.030mm | | |
| | Dielectric 1 | FR-4 | 1.030mm | 4.6 | |
| 3 | Power | CF-004 | 0.030mm | | |
| | Dielectric 3 | PP-006 | 0.203mm | 4.4 | |
| 4 | Bottom Layer | | 0.035mm | | |
| | Bottom Solder | Solder Resist | 0.012mm | 3.5 | |
| | Bottom Overlay | | | | |

Total board thickness: 1.589mm